be “SDRAM”. DDR is short for DDRSDRAM (Double Data Rate Synchronous Dynamic Random-Access Memory). The original SDRAM is now, sometimes referred to as ‘DDR1’ but DDR actually began with DDR2. If you are building a new PC, you might not have to worry about those naming conventions because both DDR1 and DDR2 modules are almost obsolete from the market. DDR4 is where all the action is right now.

DDR specifications technically deal with timings on which the chip operates. Each module has a range of operating frequencies within which it can operate with ease and it can go higher if you overclock the RAM module. The frequency determines the speed at which the RAM module can transfer data. Since RAM is the primary place the CPU looks for data after searching its own cache, speed of RAM affects the overall performance.

Then there are also RAM channels, Dual-Channel is the most popular configuration while Quad-Channel is only found on enthusiast boards. Between Dual-Channel and Single-Channel, there is a lot of performance difference but as you move to Triple Channel, that comes down a little and with Quad-Channel, the benefits are only applicable to very few applications. Simply put, the channel type dictates how many modules (or their multiple) you should have connected to ensure higher throughput. For dual channel, you need two RAM modules to be connected, for Triple-channel you need to have three modules and so on. Also, they should be of a compatible spec, it’s best if they are the same. Apart from just the frequency, latency also affects the performance. Latency is the time taken to respond to a request for data from an address within RAM. The lower the latency, the faster the RAM would work.

What RAM modules are available?

Basic Rig

HyperX FURY 4GB (HX318C10F/4) is a good performer. It is a DDR3 RAM module rated for 1866 MHz with a capacity of 4GB. Basic rigs don’t need a lot of RAM and of late, RAM has been getting a bit expensive. However, should you have surplus budget then feel free to bump this up to 8 GB.

Entry Level Gaming

HyperX FURY 4 GB 2133 MHz module from Kingston is great for a basic gaming rig. With automated overclocking, low power consumption and cool looks, it’s a pretty good module. Again, you can bump this up to 8 GB since games have become more resource heavy of late.